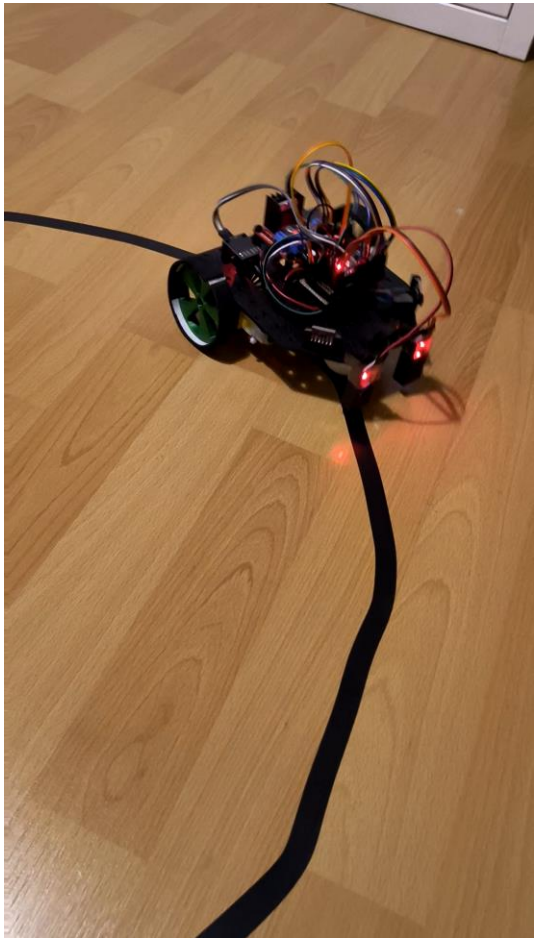
ROBOTERPROGRAMMIERUNG MIT ARDUINO

AKADEMIE DANACH 2026

WAS MACHEN WIR?

WIR BAUEN EINEN ROBO





BAUTEILE: MIKROPROZESSOR



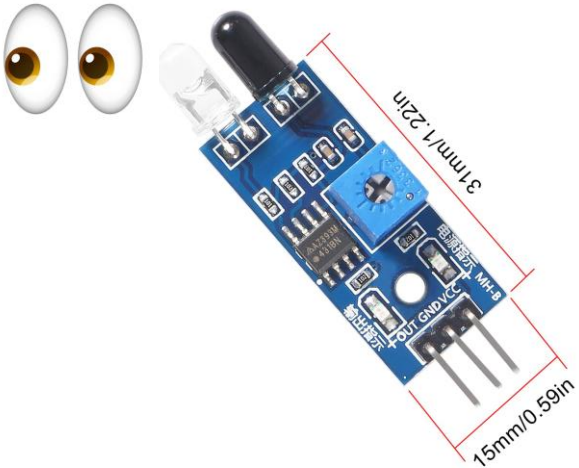


Arduino Nano

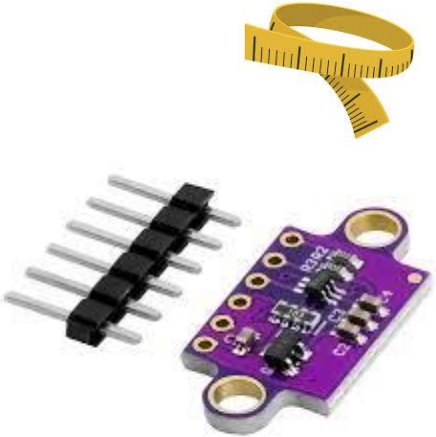




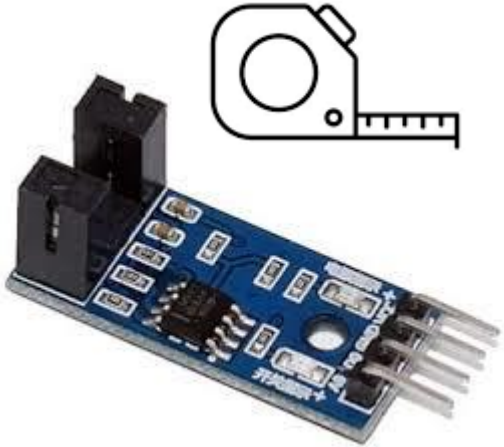
BAUTEILE: SENSOREN



Lichtreflektion



Abstand



Lichtschranke

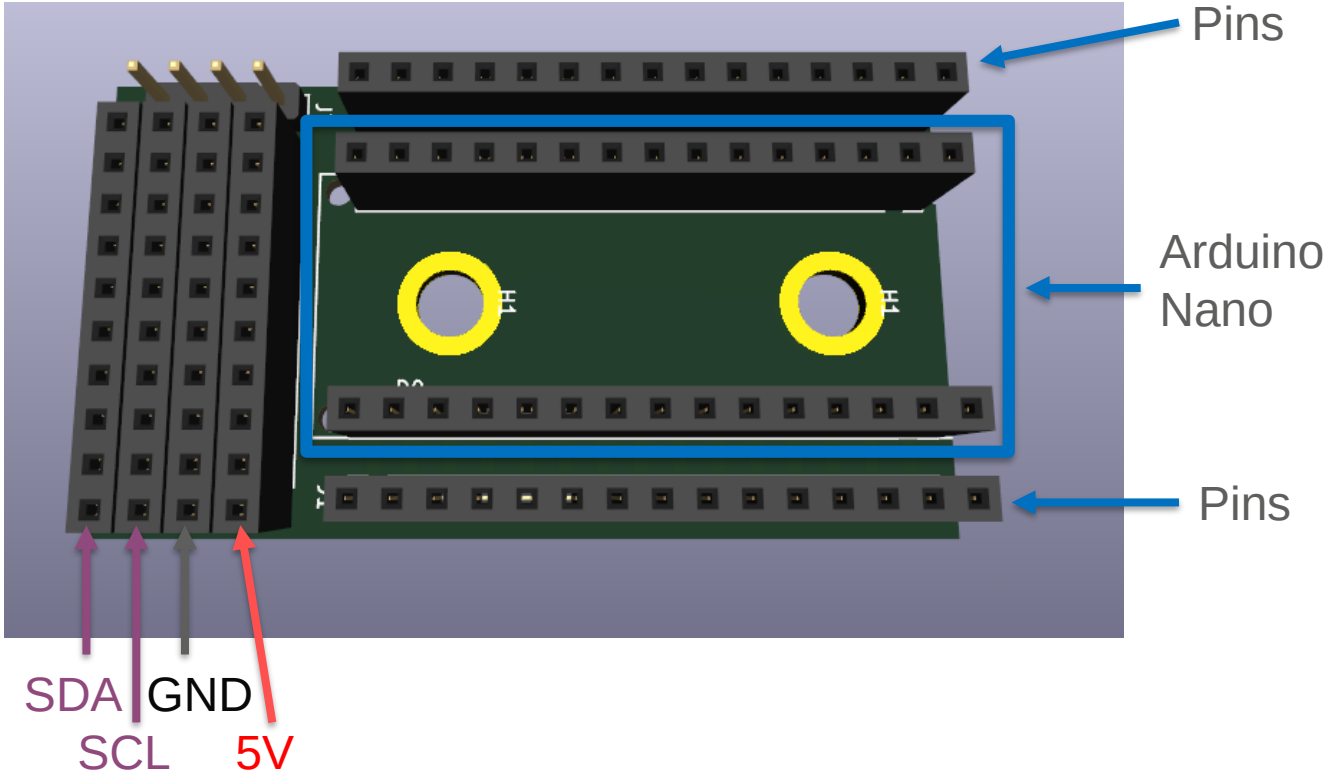
BAUTEILE: MOTOREN





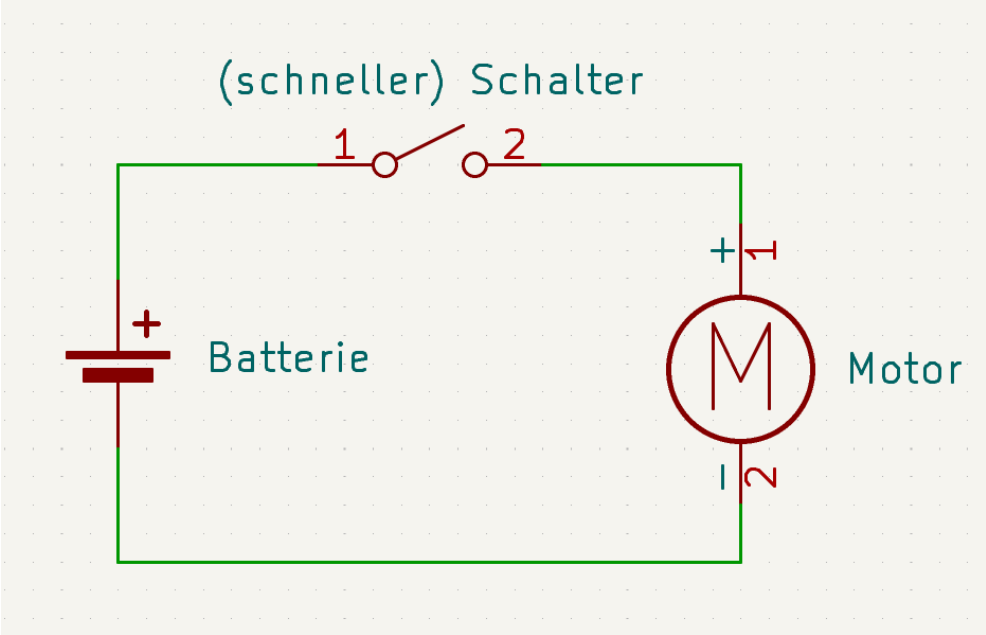


BREAKOUT-BOARD

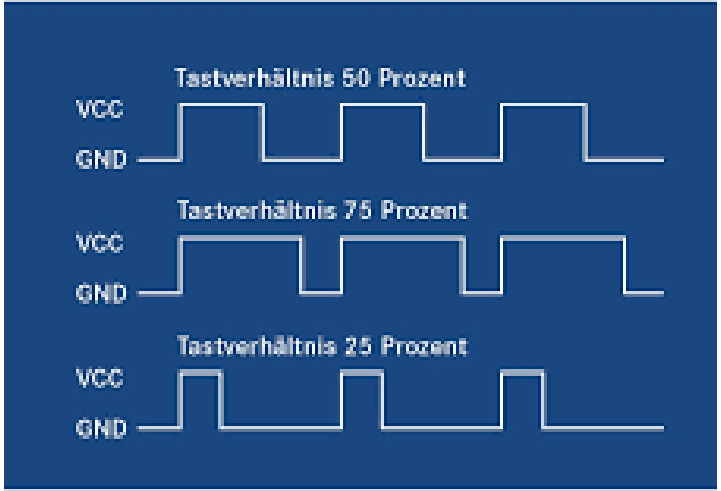


WIE FUNKTIONIERT DAS?

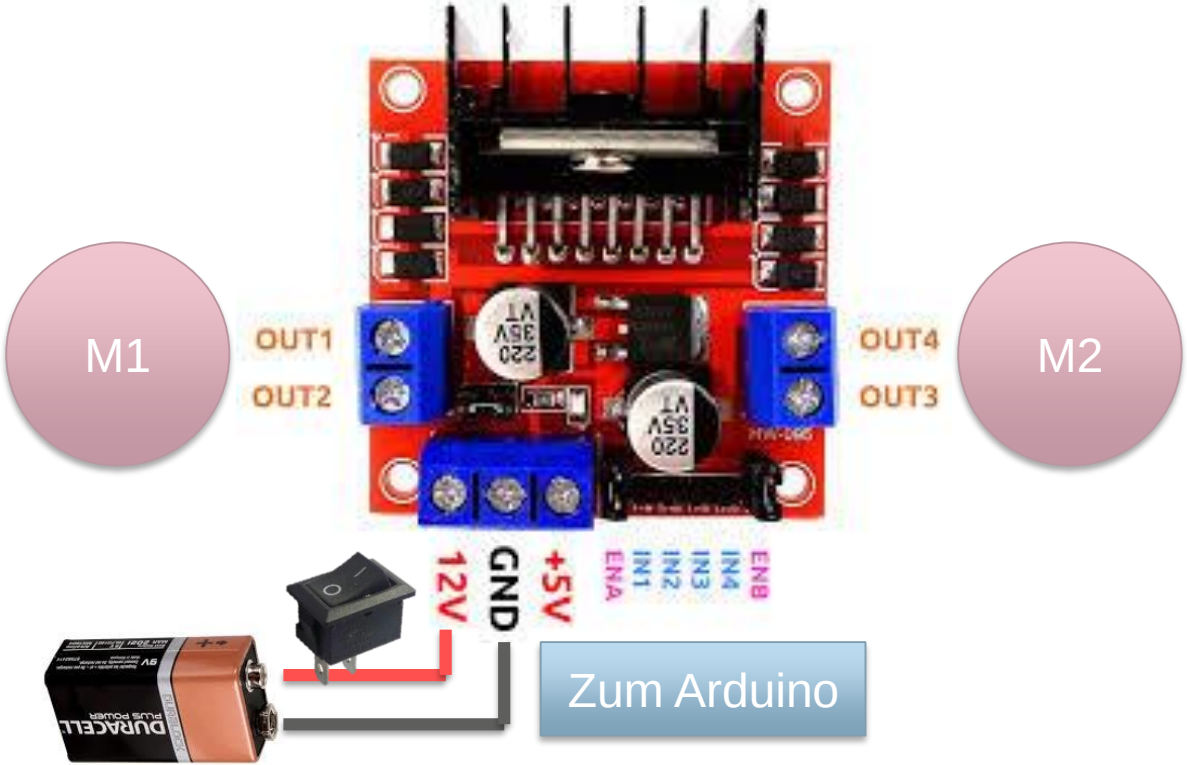
WIE BEWEGT SICH DER MOTOR?



PWM: Puls-Weiten-Modulation

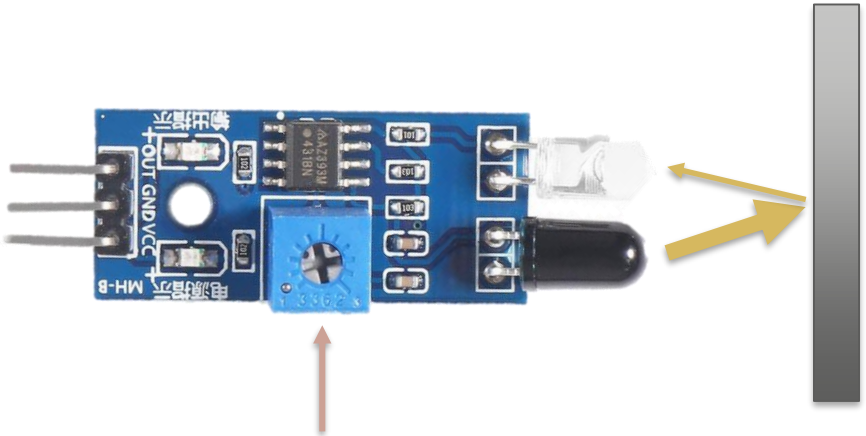


MOTOR-TREIBER



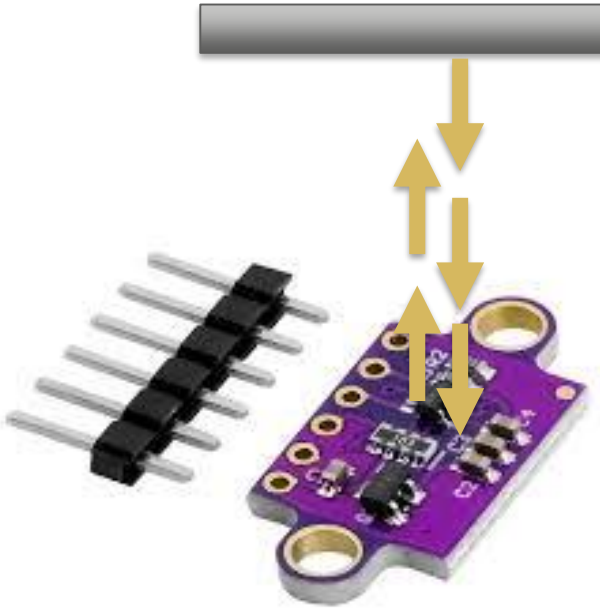
WIE FUNKTIONIEREN DIE LICHTSENSOREN?

OUT: Signal => Arduino
GND: Ground = „-“
VCC: Vers.-spannung = + 5 V



WIE FUNKTIONIEREN DIE ABSTANDSSENSOREN?

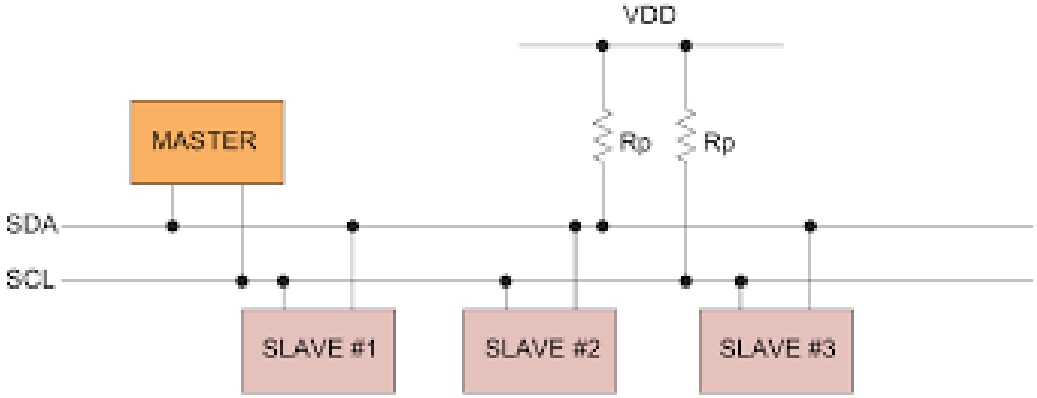
GND: Ground = „-“
VCC: Vers.-spannung = + 5 V
SCL: Serial Clock => I2C
SDA: Serial Data => I2C
XSHUT: Soft-Shutdown



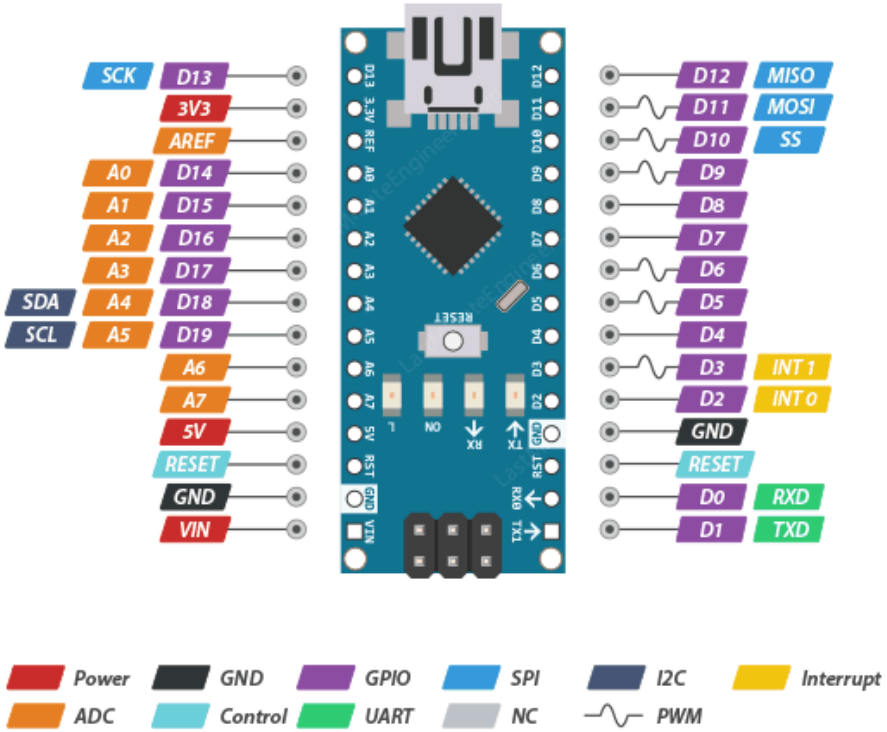
$$s = \frac{\Delta t}{c}$$

I²C: INTER-INTEGRATED-CIRCUIT BUS

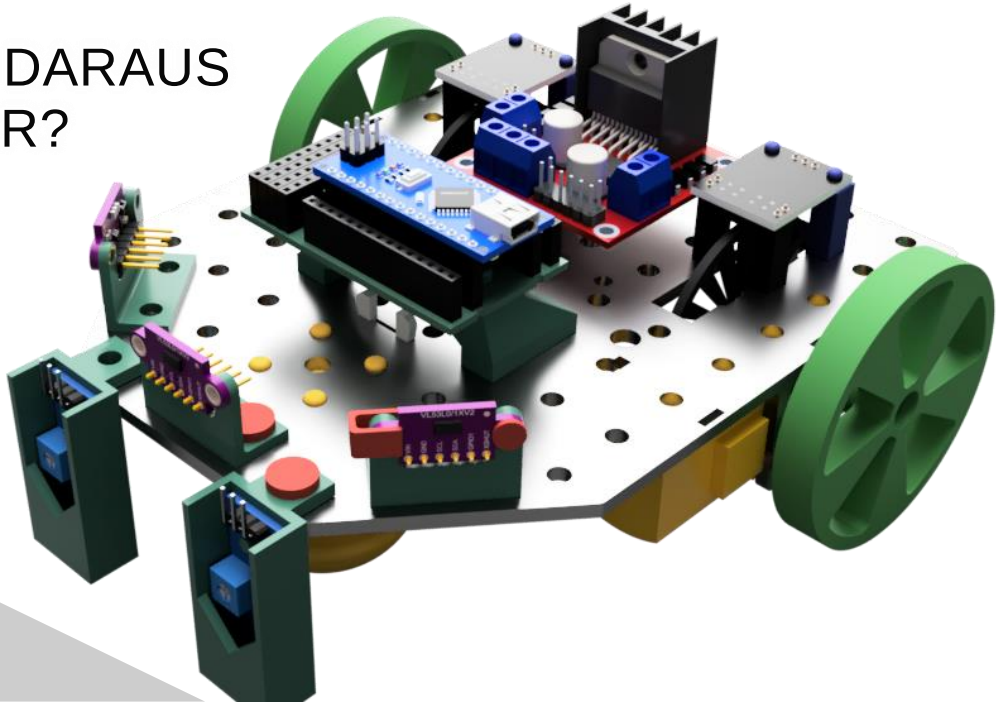
Master: Steuert alles
- Gibt Takt an (SCL)
- Sendet Daten (SDA)
- Erlaubt den anderen, zu senden



ARDUINO NANO: PINOUT

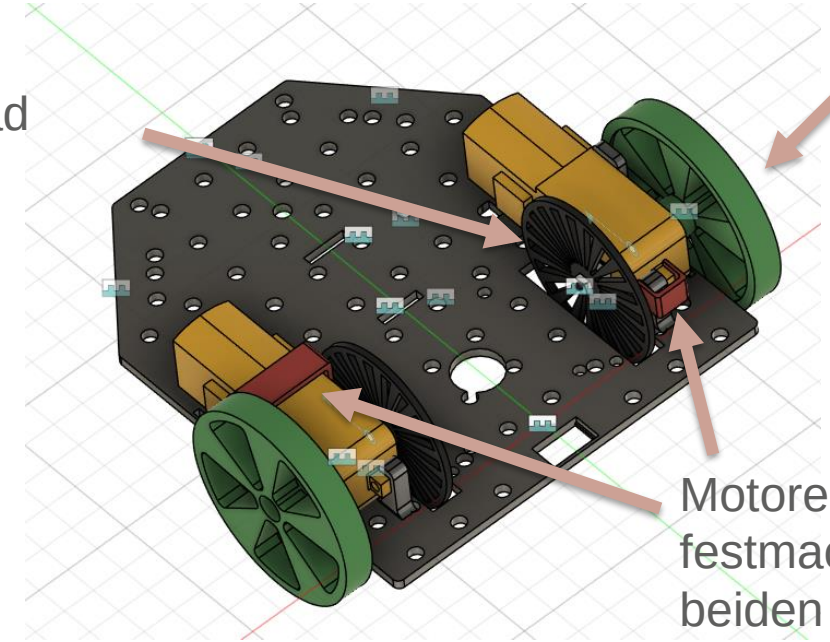


WIE BAUT MAN DARAUS EINEN ROBOTER?



BODENPLATTE UND MOTOREN

Encoderrad (später)

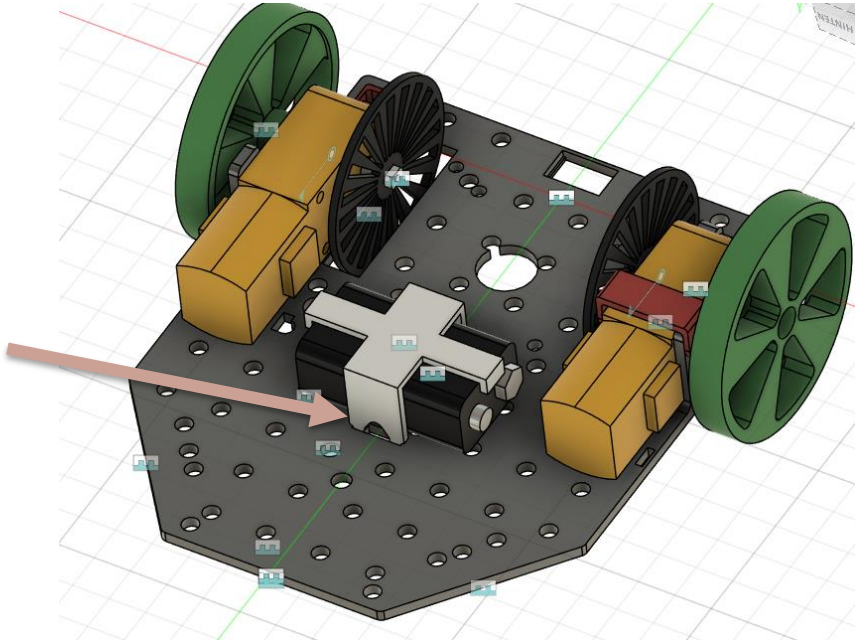


Rad

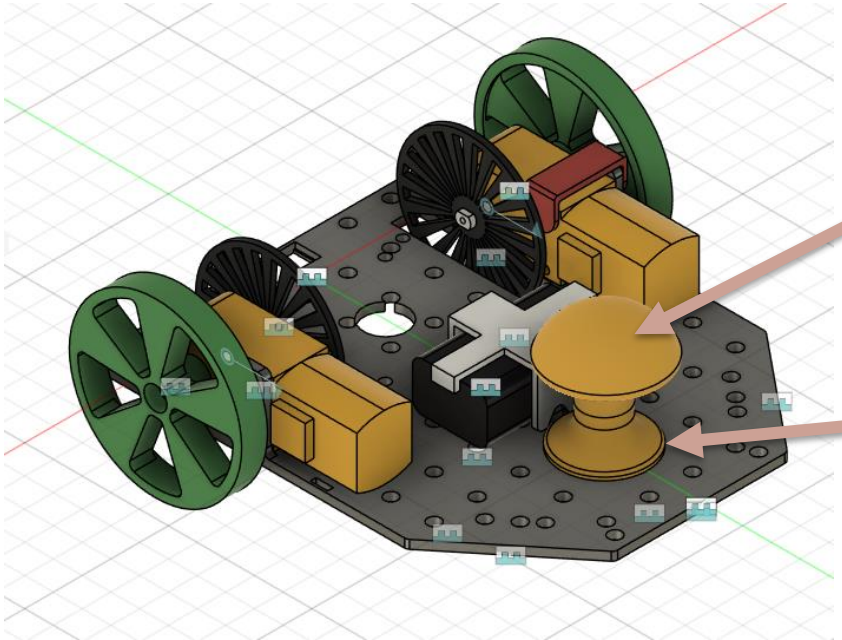
Motoren mit Clips festmachen (auf beiden Seiten)

BATTERIE

Vorsichtig in die Schlitze drücken



GLEITER („DRITTES RAD“)

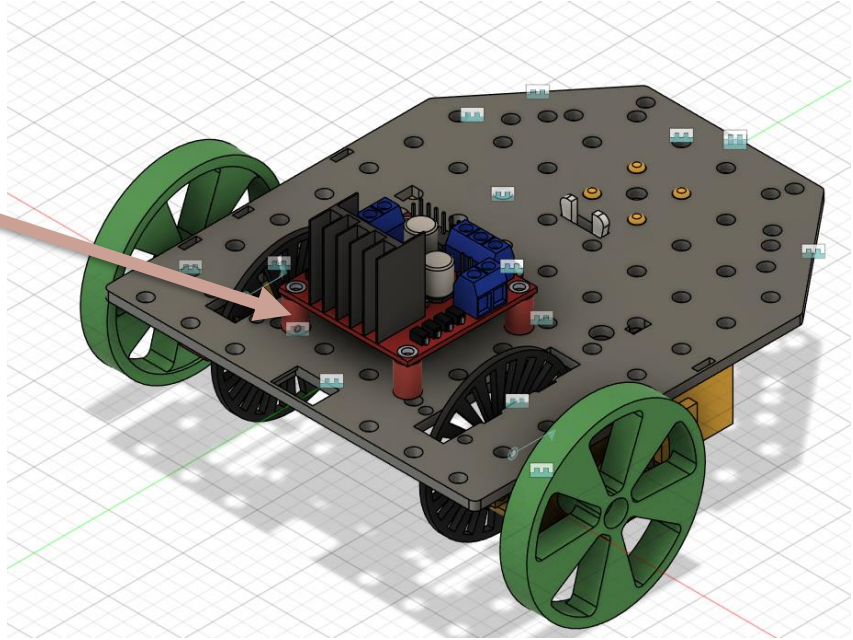


Halben Tischtennisball darauf kleben

Vorsichtig in die Löcher drücken

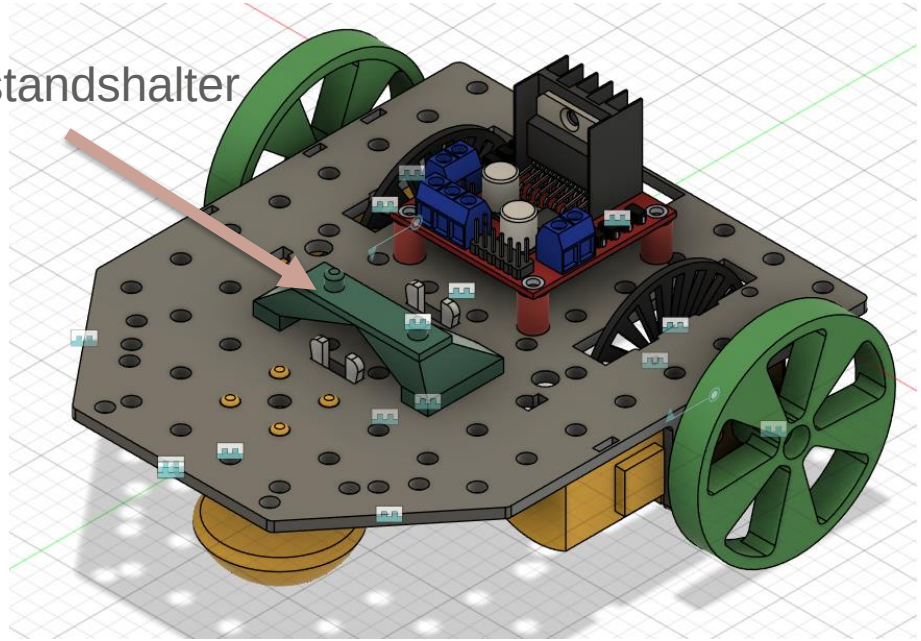
MOTORTREIBER

Mit Abstandshaltern festschrauben

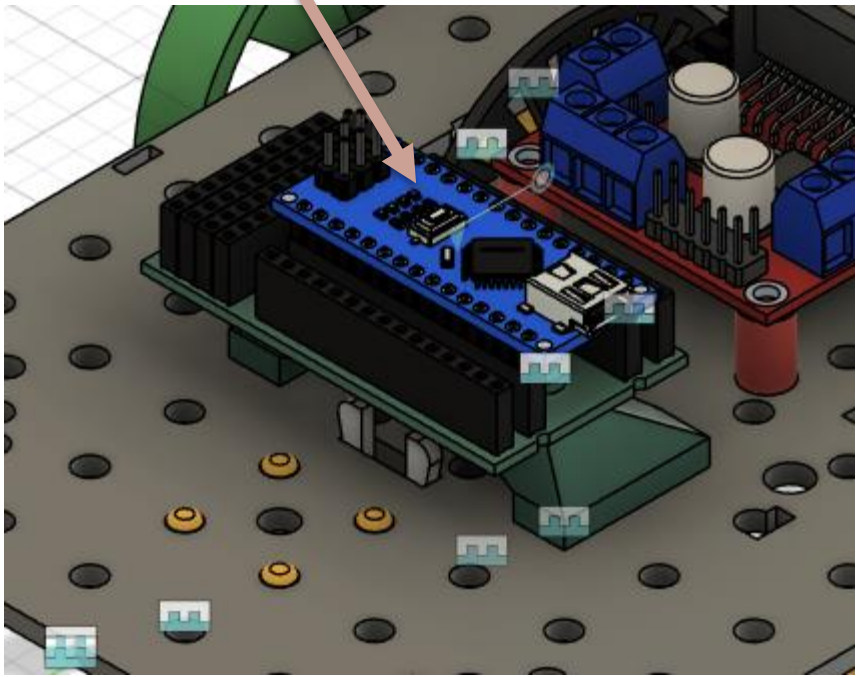


ARDUINO

Abstandshalter

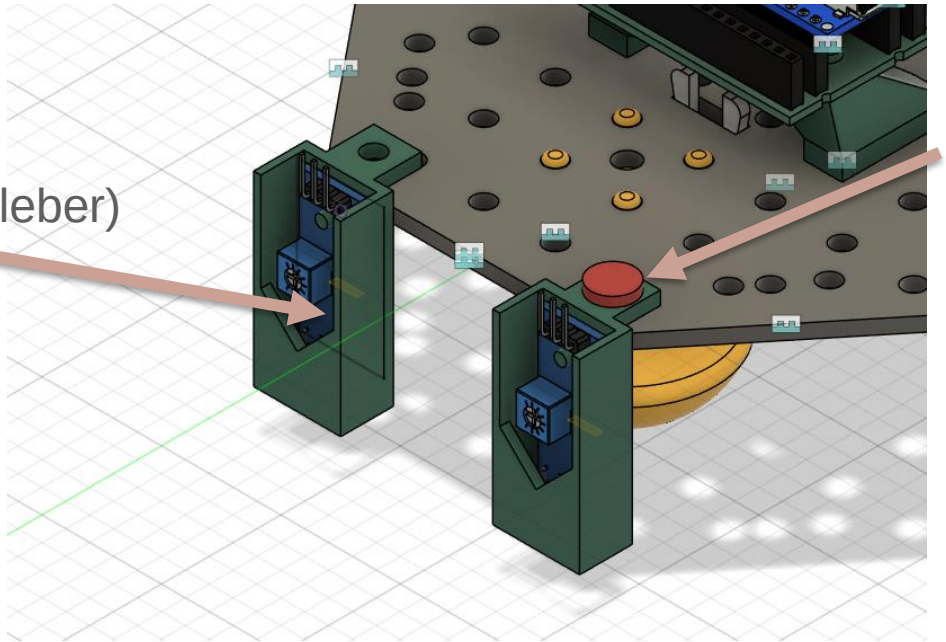


Platine + Arduino Nano



LICHTSENSOREN

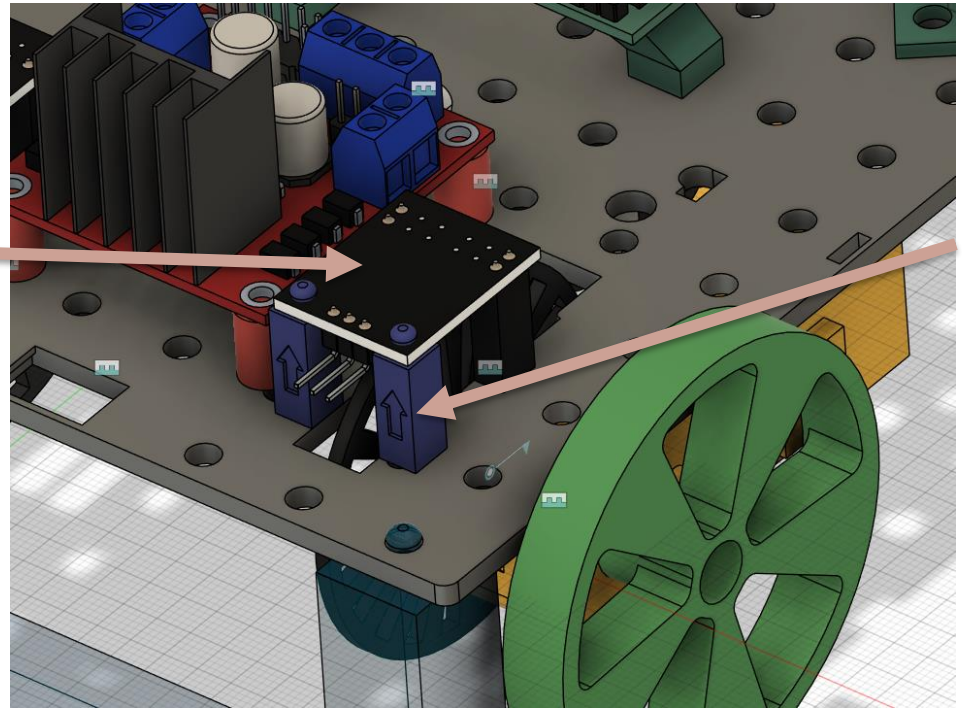
Sensor in Halter einkleben (Heißkleber)



Beide Sensoren mit Plastik-Pin vorsichtig festdrücken

MOTOR-ENCODER

Lichtschranke kopfüber über das Encoder-Rad



Vorsichtig festdrücken, eventuell kleben. Pfeilrichtung beachten

ANSCHLÜSSE

Motor-Treiber:

- 12V: Schalter – Batterie
- 5V:
  - Arduino 5V
  - I2C 5V
- GND: Alle GND zusammen
- Motoren: Links und Rechts an „Out“
- ENA => Arduino 11
- 1, 2 => Arduino 9, 10
- 3, 4 => Arduino 7, 8
- ENB => Arduino 6

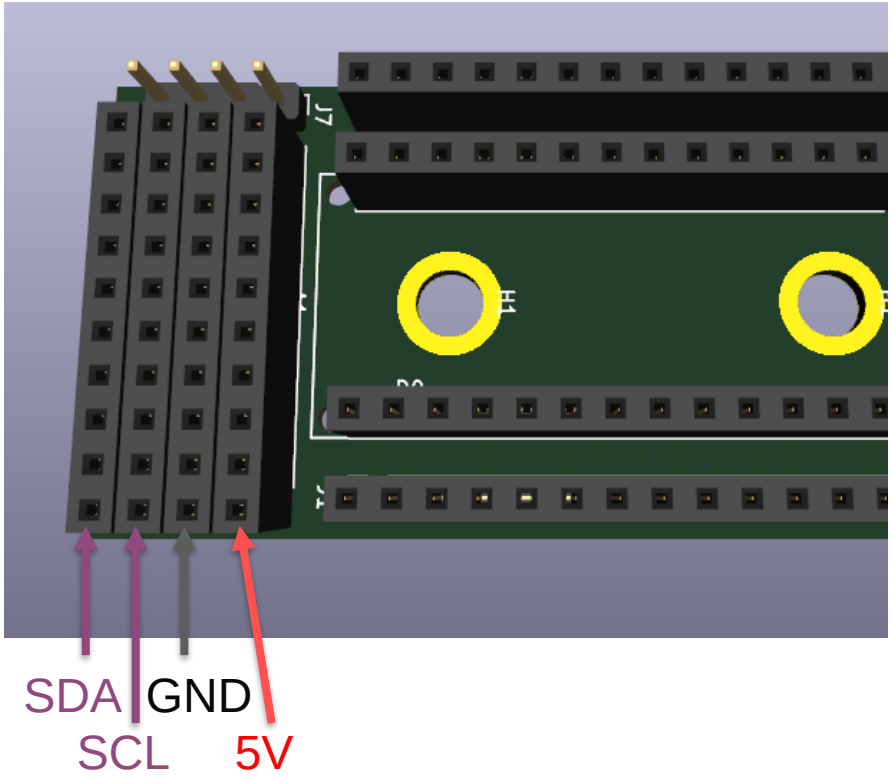
Reflektionssensor:

- VCC => 5V (Vom I2C)
- GND => GND (vom I2C)
- Out => Arduino
  - Links: Pin 4
  - Rechts: Pin 5

ANSCHLÜSSE

Abstandssensoren:

- VCC: 5V
- GND: GND
- SCL: SCL
- SDA: SDA
- XSHUT: z.B. 12 (links), 13 (rechts)

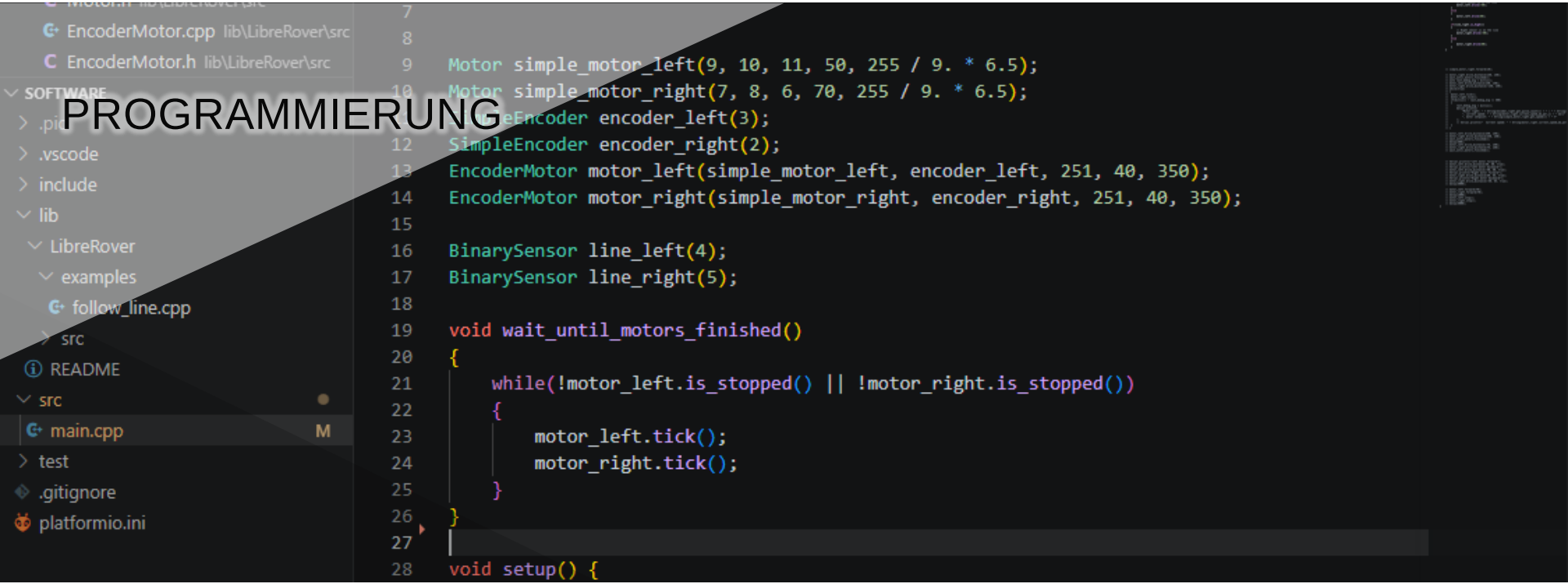


ANSCHLÜSSE

Motor-Encoder:

- VCC: 5V
- GND: GND
- OUT: 3 (links), 2 (rechts)

PROGRAMMIERUNG



PROGRAMMIERUNG 1: FAHREN

Motor *name*(
pin_forward,
pin_backward,
pin_pwm,
min_speed_pwm = 0,
max_speed_pwm = $\frac{6\,V}{9\,V} * 255$
)

```arduino
#include <Arduino.h>
#include <Motor.h>

Motor motor_left(9, 10, 11, 50, 255 / 9. * 6.5);
Motor motor_right(7, 8, 6, 70, 255 / 9. * 6.5);


void setup() {
    pinMode(LED_BUILTIN, OUTPUT);
    Serial.begin(9600);
}

void loop() {
    digitalWrite(LED_BUILTIN, HIGH);   // turn the LED on
    delay(1000);
    digitalWrite(LED_BUILTIN, LOW);    // turn the LED off

    // Turn motors on
    motor_left.forward(50);
    motor_right.forward(50);

    // Drive forward for 2 seconds
    delay(2000);

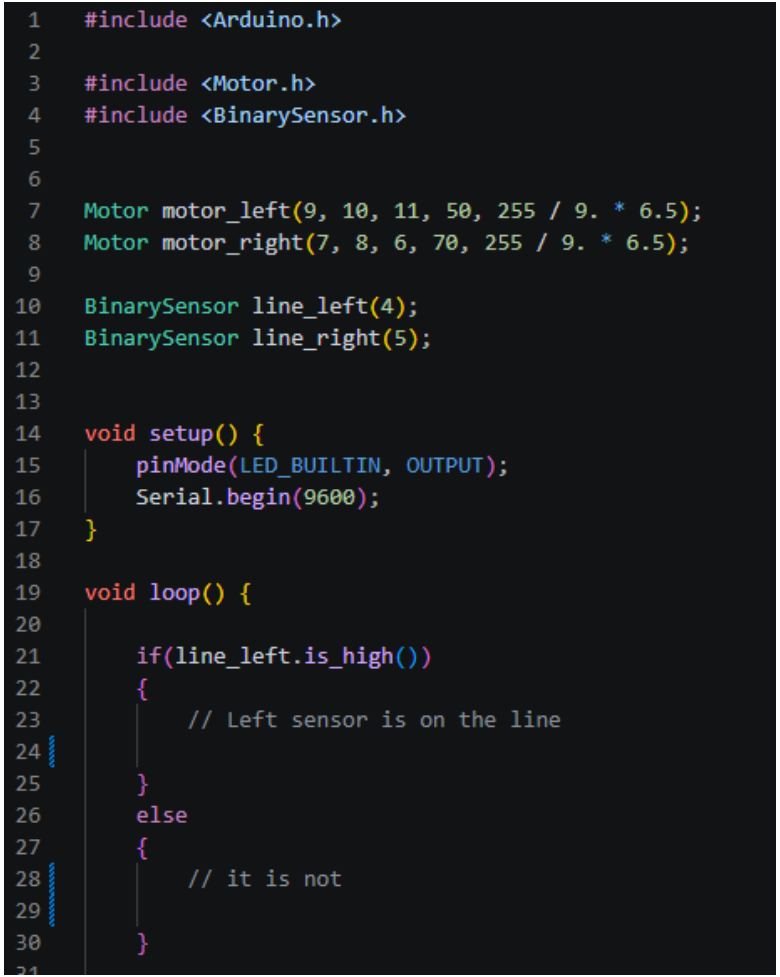
    // Turn motors off
    motor_left.stop();
    motor_right.stop();

}

```

PROGRAMMIERUNG 2: SENSOREN

```
BinarySensor name(
        pin
);
```



```
#include <Arduino.h>

#include <Motor.h>
#include <BinarySensor.h>


Motor motor_left(9, 10, 11, 50, 255 / 9. * 6.5);
Motor motor_right(7, 8, 6, 70, 255 / 9. * 6.5);

BinarySensor line_left(4);
BinarySensor line_right(5);


void setup() {
    pinMode(LED_BUILTIN, OUTPUT);
    Serial.begin(9600);
}

void loop() {

    if(line_left.is_high())
    {
        // Left sensor is on the line

    }
    else
    {
        // it is not

    }
```

DANKE!

HTTPS://GITHUB.COM/BUCHBERGERD/LIBREROVER